Introduction to

Cybersecurity

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Souvenir Photo

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Securing Accounts

Authentication

Authorization

Usernames

Passwords

Dictionary Attacks

Brute-Force Attacks

4 digits

$$10 \times 10 \times 10 \times 10$$

10^4

10,000

4 letters

$$26 \times 26 \times 26 \times 26$$

$$52 \times 52 \times 52 \times 52$$

52^4

7,311,616

4 characters

$$94 \times 94 \times 94 \times 94$$

94^4

78,074,896

8 characters

94^8

6,095,689,385,410,816

National Institute of
Standards and Technology (NIST)

"Memorized secrets SHALL be at least 8 characters in length…"

"Verifiers SHOULD permit subscriber-chosen memorized secrets at least 64 characters in length. All printing ASCII characters as well as the space character SHOULD be acceptable in memorized secrets. Unicode characters SHOULD be accepted as well."

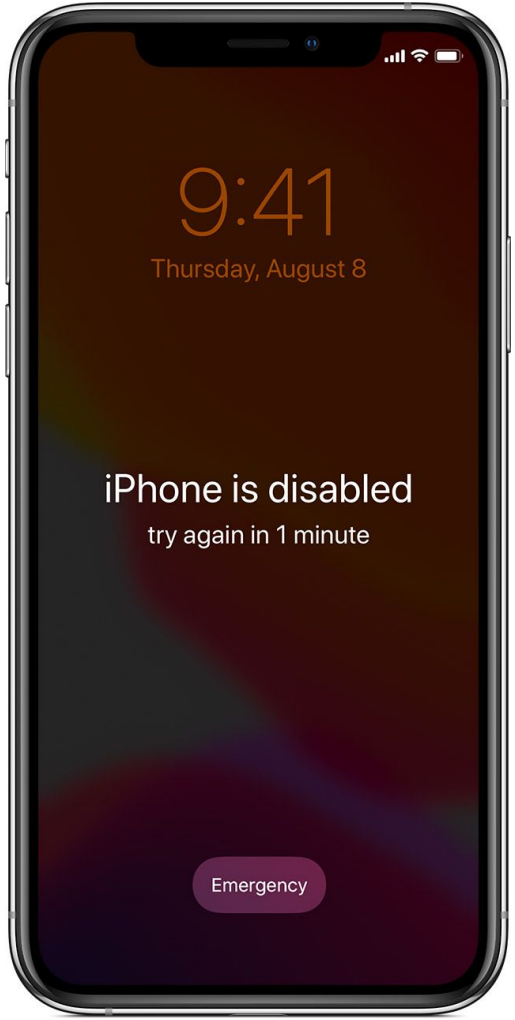
"… verifiers SHALL compare the prospective secrets against a list that contains values known to be commonly-used, expected, or compromised…

- "Passwords obtained from previous breach corpuses.
- "Dictionary words.
- "Repetitive or sequential characters (e.g. 'aaaaaa', '1234abcd').
- "Context-specific words, such as the name of the service, the username, and derivatives thereof."

"Memorized secret verifiers SHALL NOT permit the subscriber to store a "hint" that is accessible to an unauthenticated claimant. Verifiers SHALL NOT prompt subscribers to use specific types of information (e.g., "What was the name of your first pet?") when choosing memorized secrets."

"Verifiers SHOULD NOT require memorized secrets to be changed arbitrarily (e.g., periodically)."

"Verifiers SHALL implement a rate-limiting mechanism that effectively limits the number of failed authentication attempts that can be made on the subscriber’s account…"


9:41
Thursday, August 8
iPhone is disabled
try again in 1 minute
Emergency

Two-Factor Authentication (2FA)

Multi-Factor Authentication

Knowledge
Possession
Inherence
…

Biometrics

One-Time Password (OTP)

RSA
SecurID®
449 054.
Secured by RSA

123456
123456

SIM Swapping

Keylogging

Credential Stuffing

Social Engineering

Phishing

Google

Sign in

to continue to Gmail

Email or phone

Forgot email?

Not your computer? Use Guest mode to sign in privately.
Learn more

Create account

Next

Google

2-Step Verification

To help keep your account safe, Google wants to make sure it's really you trying to sign in

2-Step Verification

Get a verification code from the **Google Authenticator** app

Enter code

☑ Don't ask again on this device

Try another way

Next

Machine-in-the-Middle Attacks

Post-it®
Brand
Notes
3M

Single Sign-On (SSO)

Log in with Google

Log in with Facebook

Email

Password Forgot password?

Log in

Password Managers

Apple iCloud Keychain
Google Password Manager
Microsoft Credential Manager
...

Passkeys

"Verifiers SHALL store memorized secrets in a form that is resistant to offline attacks. Memorized secrets SHALL be salted and hashed using a suitable one-way key derivation function... Their purpose is to make each password guessing trial by an attacker who has obtained a password hash file expensive and therefore the cost of a guessing attack high or prohibitive. "

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